THE COMPARISON MORPHISM FROM THE ETALE SITE TO THE ZARISKI SITE IS FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof carries an author's footnote proving that every etale V admits an affine open cover whose members have affine open image in S; the footnote lies inside the counted span and is reproduced verbatim, and the argument it contains is a load-bearing obligation of the card.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The comparison morphism of ringed sites from the small etale site to the small Zariski site. Same chapter, Lemma lemma-compare-sites, invoked in the statement at lines 3222-3223. Supplies the morphism whose flatness is asserted and the notation for the two structure sheaves.
- Etale morphisms are open. Morphisms, Lemma morphisms-lemma-etale-open, invoked at proof lines 3231-3232. Makes the image of an etale morphism an open subscheme, which is what allows the pullback of the Zariski structure sheaf to be described by sections over that image.
- Flatness for sheaves of modules can be checked after refining a covering. Modules on Sites, Lemma sites-modules-lemma-flatness-sheafification-refined, invoked at proof lines 3251-3252. Upgrades flatness on the members of an affine open cover of each etale V to flatness of the sheaf on the whole site.

Source reference. Original lemma 070S: sources/stacks-descent/pinned-original-source.tex, source lines 2517–2517. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03WT: sources/stacks-chow/tags.txt, source lines 5124–5124. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0GN1: sources/stacks-chow/tags.txt, source lines 20776–20776. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *In Lemma 070S the morphism of ringed sites $id_{small,étale,Zar} : S_{étale} \to S_{Zar}$ is flat.*

Proof. Let us denote $\epsilon = \mathrm{id}_{small,étale,Zar}$ and $\mathcal{O}_{étale}$ and $\mathcal{O}_{Zar}$ the structure sheaves on $S_{étale}$ and S_{Zar}. We have to show that $\mathcal{O}_{étale}$ is a flat $\epsilon^{-1}\mathcal{O}_{Zar}$-module. Recall that étale morphisms are open, see Morphisms, Lemma 03WT. It follows (from the construction of pullback on sheaves) that $\epsilon^{-1}\mathcal{O}_{Zar}$ is the sheafification of the presheaf $\mathcal{O}'$ on $S_{étale}$ which sends an étale morphism $f : V \to S$ to $\mathcal{O}_S(f(V))$. If both V and $U = f(V) \subset S$ are affine, then $V \to U$ is an étale morphism of affines, hence corresponds to an étale ring map. Since étale ring maps are flat, we see that $\mathcal{O}_S(U) = \mathcal{O}'(V) \to \mathcal{O}_{étale}(V) = \mathcal{O}_V(V)$ is flat. Finally, for every étale morphism $f : V \to S$, i.e., object of $S_{étale}$, there is an affine open covering $V = \bigcup V_i$ such that $f(V_i)$ is an affine open in S for all i[1]. Thus the result by Modules on Sites, Lemma 0GN1. □

[1]Namely, for $y \in V$, we pick an affine open $y \in V' \subset V$ with $f(V')$ contained in an affine open $U \subset S$. Then we pick an affine open $f(y) \in U' \subset f(V')$. Then $V'' = f^{-1}(U') \subset V'$ is affine as it is equal to $U' \times_U V'$ and $f(V'') = U'$ is affine too.